

of the ore is somewhat different. The 16-mill composite contained 49.4 ± 0.6 dpm/g which is $95 \pm 2\%$ of the radium-226 equivalent present. The ratios are very nearly statistically identical at the 95% confidence level. Because most of the radium-226 from the original ore is known to have remained with the insoluble tailings, the same is probably equally true of protactinium. This is somewhat surprising in view of the strong complexes formed by protactinium with the sulfates and sulfuric acid used in the leaching process. Apparently, the protactinium is not dissolved.

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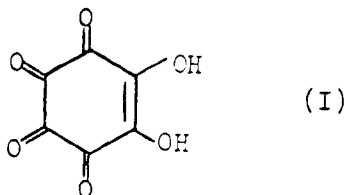
Statistical Analysis of Titration Data

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A statistically rigorous procedure is described by which equilibrium constants, the equivalence point volume, and the pH meter calibration setting are extracted from pH potentiometric titration measurements. The method takes into account the statistical uncertainties in both the pH and volume data and yields variance estimates for all the unknown parameters. The calculational procedure uses a form of iterative nonlinear regression analysis and the convergence of this iteration is facilitated by an interactive digital computer program which aids in setting initial parameter estimates. The method is illustrated by means of experimental data from the titration of dipotassium rhodizonate with standardized HCl. Rhodizonic acid ($\text{H}_2\text{C}_6\text{O}_6$) dissociation constants of $\text{p}K_1 = 4.380 \pm 0.010$ and $\text{p}K_2 = 4.659 \pm 0.004$ have been found at 25 °C.

In this paper, we describe the method by which we extract the acid dissociation constants of rhodizonic acid (I)



from pH potentiometric titration data. Rhodizonic acid is of interest because it is the six-carbon member of the series of "oxocarbon" acids which together with the corresponding anions has been studied extensively in this laboratory (1) and by West and co-workers (2). Some of the acids in this series are among the strongest known organic acids. Also, the anions are believed to be stabilized by π -electron delocalization and thus constitute a system of aromatic species of interest in quantum chemistry. In analytical chemistry, the rhodizonate anions find use as complexing agents and are used for the

determination of such metal ions as lead and calcium.

The importance of rhodizonic acid has resulted in many attempts to measure the two acid dissociation constants. The most recent measurements were made by Patton and West (3) who review the other determinations made prior to 1970, cite the wide disparities among the prior results, and explain these disparities in terms of calculational errors and of certain practical chemical difficulties encountered when working with aqueous rhodizonic acid systems. These difficulties involve an instability to decomposition in the presence of oxygen and a rather slow approach to dissociation equilibrium in certain pH ranges. Patton and West determine thermodynamic (zero ionic strength) dissociation constants of $\text{p}K_1 = 4.25$ and $\text{p}K_2 = 4.72$ for the primary and secondary dissociations at 25 °C, respectively, by a spectrophotometric method. In the present work, we have used a pH potentiometric method and our results are in rough agreement with theirs. This study is part of an extensive study of the structure of aqueous rhodizonic acid including the measurement of dissociation constants over a range of temperatures. These results will be reported in a separate paper.

MODEL EQUATIONS

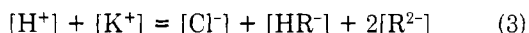
The experiments we performed involved the titration of solid samples of dipotassium rhodizonate (K_2R) dissolved in water with standardized HCl solution. At any point during the course of the titration when an aliquot of HCl has been added and after equilibrium is established, we hypothesize that species molar concentrations, dissociation constants, pH values, etc. are interrelated by a coupled set of nonlinear equations which we shall refer to as the "model equations". These are as follows: operational definition of pH

$$\text{pH} + \text{pH}_{\text{cal}} = -\log a_{\text{H}^+} = -\log [\text{H}^+] \gamma_{\text{H}^+} \quad (1)$$

conservation of rhodizonate species

$$v_{\text{ep}} F_{\text{HCl}} = (V + v) \{ [\text{H}_2\text{R}] + [\text{HR}^-] + [\text{R}^{2-}] \} \quad (2)$$

solution electroneutrality



primary dissociation equilibrium expression

$$K_1 = \frac{\gamma_{\text{H}^+} \gamma_{\text{HR}^-} [\text{H}^+] [\text{HR}^-]}{[\text{H}_2\text{R}]} \quad (4)$$

secondary dissociation equilibrium expression

$$K_2 = \frac{\gamma_{\text{H}^+} \gamma_{\text{R}^{2-}} [\text{H}^+] [\text{R}^{2-}]}{\gamma_{\text{HR}^-} [\text{HR}^-]} \quad (5)$$

stoichiometric relationships

$$[\text{K}^+] = \frac{2v_{\text{ep}} F_{\text{HCl}}}{V + v} \quad \text{and} \quad [\text{Cl}^-] = \frac{v F_{\text{HCl}}}{V + v} \quad (6)$$

semi-empirical ionic activity coefficient correlation (Debye-Hückel)

$$\log \gamma_i = \frac{-A(t) z_j^2 \sqrt{I}}{1 + B(t) d_j \sqrt{I}} \quad (7)$$

ionic strength definition

$$I = \frac{1}{2}([\text{H}^+] + [\text{K}^+] + [\text{Cl}^-] + [\text{HR}^-] + 4[\text{R}^{2-}]) \quad (8)$$

In these equations, the following notation has been used.

- a_{H^+} activity of H^+ ion
- $A(t)$, temperature dependent Debye-Hückel parameters
- $B(t)$ tabulated by Robinson and Stokes (4)
- d_j effective ion-size parameter of species j taken as 0.9 nm for H^+ and 0.6 nm for the rhodizonate anions
- F_{HCl} formal concentration of HCl solution in the buret
- I molar ionic strength
- $[j]$ molar concentration of species j
- pH_{cal} pH calibration constant
- V solution volume before adding any HCl, mL
- v volume of HCl added, mL
- v_{ep} volume of HCl required to reach the first equivalence point
- z_j electronic charge on ionic species j
- γ_j molar activity coefficient of species j

In adopting these model equations we have elected the following options.

(1) The appearance of the parameter pH_{cal} in Equation 1 represents a calculational method of checking the calibration setting of the pH meter. This is advantageous when titrating a base with HCl because once the second equivalence point is past, the solution consists of HCl at an ionic strength slightly enhanced by K^+ and Cl^- . The pH of such solutions are reliably known at ionic strengths below about 0.1 M and, hence, these may serve as calibrating solutions for the meter. This method avoids the experimental uncertainty which is introduced by the conventional method of calibrating with a buffer solution, withdrawing, rinsing, and drying the electrodes, and then inserting them in the titration solution which presents a different geometrical and electrical environment than the buffer.

(2) The effect of OH^- is neglected in Equations 3 and 8 because the recorded pH values never exceed 5.6.

(3) Additivity of volumes v and V is assumed. Also, the volume increment due to addition of approximately 50 mg of solid K_2R is neglected.

(4) We have chosen to determine the quantity of rhodizonate in solution from the equivalence point volume and the standardized HCl concentration rather than from the weight of solid K_2R . This is done because the purity of the solid is not ensured. We have also assumed that any impurity in the K_2R sample is essentially inert as an acid or base and that its effect on the ionic strength is negligible.

All quantities in these equations will be classified into three categories: parameters, constants, and variables. The parameters are the unknown values which we seek: K_1 , K_2 , v_{ep} , and pH_{cal} . The constants are both experimental, F_{HCl} and V , and semi-empirical, $A(t)$, $B(t)$, and d_j . The variables are all the remaining quantities whose values change during the course of the titration. We now must distinguish between the variables v and pH which are elements of the model equations and the measurements v_d and pH_d which are the data points. The quantities v and pH are calculable by means of the model equations and a plot of pH vs. v is the calculated titration curve. v_d differs from v and pH_d differs from pH because of the inevitable random errors of measurement and possibly because of discrepancies between the model equations and "reality". These latter discrepancies are the systematic or determinate errors. Because of these random and systematic errors, it is incorrect to make only four titration readings which is the minimum number required to determine the four unknown parameters. Also we require more information than simply four unknown values and that is statistical uncertainties of these unknowns and some indication of systematic error, if any exists. Consequently, we made over 30 measurements during the titration. In the next section, we explain the rationale and outline the methodology of the calculational procedure which uses these measurements to yield the desired parameter values along with their statistical uncertainties as well as some indication of possible systematic errors.

CALCULATIONAL METHOD

Unknown parameter values are calculated from experimental data by least-squares regression procedures. Usually, these procedures are facilitated by experimental situations in which the precision of one data set is considerably less than the other, and in which the less precise data values (the set denoted as the dependent variable) have a uniform variance over the course of the experiment. However, we shall now show that these conditions do not properly apply to the rhodizonate titration reported here.

Our experimental measurement of pH was done with a digital pH meter which reads to 0.001 pH unit. Considering also the fact that electrodes were not removed from the solution during the course of the titration, we estimate that the statistical uncertainty of each pH_d value which is due only to electronic circuit noise and turbulence effects of the magnetic stirring apparatus and estimate this uncertainty to be approximately ± 0.002 pH unit. However, in this particular titration there is an additional source of statistical uncertainty and that is due to the slow approach to chemical equilibrium in the solution as mentioned earlier. Although 15 to 30 min were allowed for each solution to equilibrate, and although the pH meter readings were unchanging for at least 3 min before the values were recorded, we could not be absolutely certain that the eventual equilibrium values were reached. Deviations from equilibrium would tend to scatter statistically and thus comprise another source of statistical uncertainty in the pH_d values. Considering that we noticed that the approach to equilibrium before the first equivalence point was slower than after this point and that after the second equivalence point there is no further rhodizonate reaction, we estimate the following overall statistical uncertainties apply to pH_d values in particular data ranges: ± 0.005 pH unit for $0 < v < v_{\text{ep}}$; ± 0.003 pH unit for $v_{\text{ep}} < v < 2v_{\text{ep}}$; and ± 0.002 pH unit for $v > 2v_{\text{ep}}$. A rough average value is ± 0.003 pH unit over the whole titration, and since the average pH_d reading was ~ 3 , these data have an estimated statistical uncertainty of about 0.1%. The buret used to deliver aliquots of HCl solution was a 10-mL microburet with 0.05-mL divisions. By interpolating between rulings, we recorded v_d values to three decimal places and estimate statistical uncertainties of ± 0.005

mL due to reading error. Considering that the v_d values averaged 4 mL, their statistical uncertainties were also about 0.1%. Since neither of these variables has been measured with significantly higher precision than the other, it is not obvious which variable is to be regarded as independent and which is to be dependent. The situation is further complicated by the fact that the titration curve has a variable slope. In the neighborhood of the second equivalence point the absolute magnitude of the slope of the pH vs. v curve is greater than elsewhere. Consequently, in this region, the uncertainty in pH_d carries less importance than the uncertainty in v_d . The relative importance of pH_d vs. v_d measurements depends on the local slope which varies. These factors require that any calculational procedure used to fit the model equations to the observed data must regard both sets of observations as having statistical uncertainty, and that each data point be weighted by a function of the local slope. These factors can be treated properly by a method due to Deming (5) which has been well described in the chemistry literature by Wentworth (6). Considering the ready availability of these publications, the following exposition will be limited to comments on our adaption of Deming's procedure to this particular calculation. Much of the notation in this section will be the same as that used by Deming and Wentworth.

The Condition Equation. There is a single "condition equation" which could in principle be written down by combining all the model equations 1-8 together by substituting and eliminating all the variables (not the quantities we have called parameters and constants) in favor of pH and v . In practice though, these nonlinear equations cannot be reduced to a single equation relating pH, v , and all the parameters and constants. They can, however, be solved numerically and since the "condition equation" need not be so reduced, we identify the full set of model equations as Deming's "condition equation." This entity $F(x, y; a, b, c) = 0$ is interpreted as follows.

1. Arbitrarily identify y with v and x with pH. This choice is made because it is a simpler matter to solve the model equations for v given a pH value than vice-versa. How this is actually accomplished is described below in a section entitled "Evaluation of F_0 and Derivatives".

2. Identify the parameters $a, b, c, \dots$ with K_1 or $\text{p}K_1, K_2$ or $\text{p}K_2, v_{\text{ep}}$, and pH_{cal} .

3. Let the numerical solution of the model equations for v given values of parameters, constants, and pH be denoted by $y = v(x; a, b, c)$. The "condition equation" is

$$F(x, y; a, b, c) = v(x; a, b, c) - v_d = 0 \quad (9)$$

and the "condition function" F is $v(x; a, b, c) - v_d$. This form of F is the negative of the form used by Deming and this modification is made so that derivatives of F with respect to parameters shall have the same algebraic sign as the derivatives of v with respect to the same parameters.

Least-Squares Minimization. The calculational procedure minimizes the weighted sum-of-squares S of both residuals V_{xi} and V_{yi} over all n recorded points.

$$S = \sum_i (w_{xi} V_{xi}^2 + w_{yi} V_{yi}^2) \quad (10)$$

subject to the restraint imposed by the condition equation. The residuals are defined here as

$$V_x = \text{pH}_d - \text{pH} \text{ and } V_y = v_d - v \quad (11)$$

and the weighting factors w_x and w_y are inversely proportional to σ_x^2 and σ_y^2 , respectively. These latter quantities are the a priori variance estimates of the x and y measurements. The minimization of S is done iteratively by choosing a set of approximate parameter values a_0, b_0, c_0 which are reasonably close to the true a, b, c and then finding corrections $\Delta a, \Delta b,$

Δc to a_0, b_0, c_0 . The refined set of parameters $(a_0 - \Delta a), (b_0 - \Delta b), (c_0 - \Delta c)$ then are used as approximate values for another iteration and another refinement and so on. The corrections are found by solving a set of linear algebraic equations for the unknowns $\Delta a, \Delta b, \Delta c$ and this is done by a matrix inversion method. The elements that must be calculated are

$$\sum_{i=1} \frac{\left(\frac{\partial F}{\partial a}\right)_i \left(\frac{\partial F}{\partial b}\right)_i}{L_i}$$

for a square $\mathbf{p} \times \mathbf{p}$ matrix and

$$\sum_{i=1} \frac{F_{0i} \left(\frac{\partial F}{\partial a}\right)_i}{L_i}$$

for a p component vector. Here p is the number of parameters being refined and

$$L_i \equiv \frac{1}{W_i} = \frac{\sigma_x^2}{\sigma^2} \left(\frac{\partial F}{\partial x}\right)_i^2 + \frac{\sigma_y^2}{\sigma^2} \left(\frac{\partial F}{\partial y}\right)_i^2 \quad (12)$$

W_i is interpreted as the weighting factor for data point i and is seen to depend on the measurement variances $\sigma_x^2 = \sigma^2/w_x$ and $\sigma_y^2 = \sigma^2/w_y$ and on the local slope of the condition function. σ^2 is an arbitrary constant of proportionality. F_{0i} is the condition function evaluated with variables and parameters initially taken as $\text{pH}_{di}, v_{di}, a_0, b_0, c_0$ and later by their refinements when these have been calculated. $(\partial F/\partial a)_i$ and $(\partial F/\partial x)_i$ are partial derivatives of F evaluated similarly.

When the iterative refinement of parameters has converged, other useful information is available. The x -residuals at each point are calculated from

$$V_{xi} = \frac{W_i}{w_x} F_{0i} \left(\frac{\partial F}{\partial x}\right)_i \quad (13)$$

and the V_{yi} are calculated similarly. F_{0i} is now the condition function at point i but with the refined parameters substituted. From these residuals, it is a simple matter to use Equation 11 to calculate pH_i and v_i which are the coordinates of a point on the calculated titration curve.

The quality of the fit of this calculated curve to the data points is reflected in the sum S of Equation 10. This sum could be interpreted directly as the sum of squared deviations were it not for the weighting factors w_x and w_y which are inversely proportional to σ_x^2 and σ_y^2 , respectively, but otherwise arbitrary in magnitude. We have chosen to depart slightly from Deming's convention here by using his arbitrary multiplier σ^2 to normalize the weights W_i to an average value of unity. In so doing we define an average weighting factor $\bar{W}$ by

$$n\bar{W} \equiv \sum_i W_i = \sum_i \left[\sigma_x^2 \left(\frac{\partial F}{\partial x}\right)_i^2 + \sigma_y^2 \left(\frac{\partial F}{\partial y}\right)_i^2 \right]^{-1} \quad (14)$$

and then divide each W_i, w_x , and w_y by $\bar{W}$ to yield normalized weighting factors which will be denoted by asterisks. The normalized W_i^* then becomes

$$W_i^* = \frac{1}{\bar{W}} \left[\sigma_x^2 \left(\frac{\partial F}{\partial x}\right)_i^2 + \sigma_y^2 \left(\frac{\partial F}{\partial y}\right)_i^2 \right]^{-1} \quad (15)$$

which when compared with Equation 12, shows that σ^2 is identified with $1/\bar{W}$. With this normalization, the weighted sum-of-squared residuals becomes

$$S^* = \sum_i (w_x^* V_x^2 + w_y^* V_y^2) = \sum_i W_i^* F_{0i}^2 \quad (16)$$

The latter equation results from substituting Equation 13 for

the residuals and recognizing the identity written as Equation 15. When F_{0i} in Equation 16 is evaluated with refined parameters, then this quantity is the deviation of the data point v_{di} from the calculated titration curve as seen by Equation 9 and W_i^* is a weighting factor whose average value is unity over the course of the titration. If the weighted sum-of-squared deviations represented by Equation 16 is divided by the number of degrees of freedom, the result is the variance of residuals s_r^2 :

$$s_r^2 = \frac{S^*}{n - p} \quad (17)$$

which is the same as Deming's σ_{ext}^2 . This quantity should approximate the effective variance of the measurements if systematic error (lack-of-fit) does not contribute to the deviation of the calculated curve from the data points.

However, because we have a priori estimates of the statistical uncertainties σ_y^2 and σ_x^2 of the v_d and pH_d measurements, respectively, we have a means of estimating the a priori effective variance of the measurements. We shall denote this a priori variance by s_m^2 and shall take this to be

$$s_m^2 = \frac{n}{\sum_{i=1}^n \left[\sigma_x^2 \left(\frac{\partial F}{\partial x} \right)_i^2 + \sigma_y^2 \left(\frac{\partial F}{\partial y} \right)_i^2 \right]^{-1}} \quad (18)$$

which when compared with Equation 14 shows that s_m^2 identifies with Deming's σ^2 as both these quantities equal $1/\bar{W}$. The consistency of this formulation is checked by observing that $\chi^2 = S^*/s_m^2$ and that by taking the ratio of Equations 16 and 18 and substituting for $\bar{W}$, χ^2 turns out to be $\sum_i ((V_{xi}^2/\sigma_x^2) + (V_{yi}^2/\sigma_y^2))$ which is properly the dimensionless sum-of-squared residuals. This quantity can be used to test the randomness of the residuals.

Finally, the diagonal elements of the inverse of the $\mathbf{p} \times \mathbf{p}$ matrix are the variance coefficients of the parameters, i.e., if c_{22} is such an element corresponding to parameter b , then the variance $\text{var}(b) = s^2 c_{22}$. s^2 may be estimated from a priori information as s_m^2 or by s_r^2 provided that systematic error is negligible. These variances may serve as statistical uncertainty estimates of the parameters.

Evaluation of F_0 and Derivatives. The condition function F and its derivatives are calculated numerically from the model equations by the following strategies. All these quantities are calculated at each recorded titration point.

The concentration of all rhodizonate species are eliminated from the model equations by substituting $[\text{R}^{2-}]$ of Equation 5 and $[\text{HR}^-]$ of Equation 4 into the rhodizonate conservation Equation 2 yielding

$$v_{\text{ep}} F_{\text{HCl}} = (V + v)[\text{H}_2\text{R}]P_1 \quad (19)$$

and into the electroneutrality Equation 3 yielding

$$[\text{H}^+] + \frac{2v_{\text{ep}} F_{\text{HCl}}}{V + v} = \frac{v F_{\text{HCl}}}{V + v} + [\text{H}_2\text{R}]P_2 \quad (20)$$

Here $P_1 = (1 + (K_1/G_1 a_{\text{H}^+}) + (K_1 K_2/G_1 G_2 a_{\text{H}^+}^2))$; $P_2 = ((K_1/G_1 a_{\text{H}^+}) + (2K_1 K_2/G_1 G_2 a_{\text{H}^+}^2))$; $G_1 = \gamma_{\text{H}^+} \gamma_{\text{HR}^-}$; and $G_2 = \gamma_{\text{H}^+} \gamma_{\text{R}^{2-}} / \gamma_{\text{HR}^-}$.

Then $[\text{H}_2\text{R}]$ is eliminated between Equations 19 and 20 and the result rearranged to solve for v .

$$v = \frac{\left(2 - \frac{P_2}{P_1} \right) v_{\text{ep}} F_{\text{HCl}} + [\text{H}^+] V}{F_{\text{HCl}} - [\text{H}^+]} \quad (21)$$

To obtain the numerical solution for F_0 we have chosen to embed two other iterations within the overall titration seeking a , b , c , ... from a_0 , b_0 , c_0 . One of these which we shall call the

" G iteration" initializes with approximate guesses of $\log \gamma_{\text{H}^+}$ and $\log \gamma_{\text{HR}^-}$, the other, called the " v iteration," initializes with guesses of v at each point. The quantities $\log \gamma_{\text{H}^+}$, $\log \gamma_{\text{HR}^-}$ and v are refined by successive substitutions in the model equations until convergence criteria are met simultaneously for all three variables. We can report that these equations are well-behaved and so there were no convergence problems encountered. Numerical evaluation of F_0 proceeds as follows at each data point.

1. Calculate $\text{pH}' = \text{pH}_d + \text{pH}_{\text{cal}}$ and $a_{\text{H}^+} = 10^{-\text{pH}'}$ where pH_{cal} is one of the parameter guesses.
2. Initialize the G iteration with $\log \gamma_{\text{H}^+} = \log \gamma_{\text{HR}^-} = 0.1$.
3. Initialize the v iteration with $v = v_d$.
4. Calculate $[\text{H}^+] = a_{\text{H}^+} / \gamma_{\text{H}^+}$.
5. Calculate G_1 , G_2 , P_1 , and P_2 using Equation 7 and the parameter guesses of K_1 and K_2 .
6. Calculate $[\text{Cl}^-]$ and $[\text{K}^+]$ from Equations 6.
7. Calculate $[\text{HR}^-]$ from the conjunction of Equations 3 and 5 having eliminated $[\text{R}^{2-}]$ from these two.
8. Calculate $[\text{R}^{2-}]$ from Equation 5.
9. Calculate I from Equation 8.
10. Calculate refined $\log \gamma_{\text{H}^+}$ and $\log \gamma_{\text{HR}^-}$ from Equation 7.

11. Check the G iteration for successful convergence to $\log \gamma_{\text{H}^+}$ and $\log \gamma_{\text{HR}^-}$ to ± 0.001 log unit. If unsuccessful, repeat steps 4–11.

12. Calculate a refined v from Equation 21.

13. Check the v iteration for convergence to $\sigma_v/10 = \pm 0.0005$ mL. If unsuccessful, repeat steps 4–13 using the refined v .

14. Check the simultaneity of convergence of both iterations by requiring both checks at steps 11 and 13 to be successful without further refinement of any of the three check variables. If unsuccessful, repeat steps 4–14 until successful.

15. Calculate $F_0 = v - v_d$.

No attempt was made to find closed-form expressions for derivatives of F . Derivatives with respect to the parameters were evaluated numerically by calculating an incremental change dF_0 in F_0 due to an incremental change dp in p and taking the ratio dF_0/dp . The incremental change dp was arbitrarily chosen as $(p + 1) \times 10^{-5}$.

The derivative $(\partial F/\partial y)_i$ is $(\partial F/\partial v_d)_i$ which is -1 for all i . $(\partial F/\partial x)_i$ is evaluated as was done for the parameter derivatives by finding incremental changes in F due to incremental changes in pH .

INTERACTIVE DIGITAL COMPUTER PROGRAM

It is well-known that the success or failure of convergence of the parameter refinement iteration depends critically on the proximity of the initial parameter estimates to those optimal parameter values which minimize S . Deming's method, the basis of which was originated by Gauss, is essentially a linearization of the nonlinear model equations around each parameter value and this linearization can be accurate only in the immediate neighborhood of the optimal values. We have developed an interactive computer program which offers the operator some flexibility in the preliminary stages of the calculation before the parameter refinement iteration begins. By utilizing certain options, he can adjust and readjust parameter values until satisfied that these are close enough to the optimal values to attempt the iteration. If the iteration should then diverge or should it converge to a spurious set of values corresponding, perhaps, to a spurious local minimum in S , the parameters can be readjusted again and another iteration attempted. The program's options are as follows.

1. Reset any or all parameter values.

2. Calculate the "fit", i.e., the rms weighted deviation of the data points from the calculated titration curve. This

quantity, which is proportional to S , serves to monitor the quality of the current set of parameter values.

3. Holding all parameters fixed save one, vary this one parameter over a selected range of values and calculate the fit over this range. This feature allows the operator to scan the fit with respect to variation in any parameter and thereby explore the S surface for the direction of the minimum.

4. Iterate for the refinement of any number of parameters. Those parameters which are not being refined are held constant. The iteration proceeds for a preselected number of steps and the fit is printed out at each step. The operator observes the decrease of fit and, when the preselected steps have been taken, he may opt to continue iterating or not depending on the rate of decrease of successive fits. How each step is actually taken is described in more detail below.

5. Restrict the range of data points over which the fit or the parameter iteration is calculated. This feature is particularly useful in this titration problem because certain parameters are particularly sensitive to the data in particular ranges. For example, the primary dissociation K_1 is a sensitive function of the data only when $[HR^-]$ and $[H_2R]$ are both significant in solution together. This occurs between the first and second equivalence points when K_2R is titrated with HCl. Similarly K_2 is a sensitive function of the data only prior to the first equivalence point. Also neither K_1 nor K_2 are important beyond the second equivalence point, but pH_{cal} depends critically on the data in that region.

6. Print out all the data values, the residuals, and the calculated pH and ν values.

7. Print out a statistical summary including S^* , s_r^2 , s_m^2 , χ^2 , and the variance coefficients of the parameters.

It is well-known that in certain cases involving highly nonlinear model equations and sufficiently poor initial parameter guesses a_0 , b_0 , c_0 , that refinement of, for example, a_0 to $(a_0 - \Delta a)$ diverges from or converges very slowly to the optimal value a . Bard (7) reviews many alternative methods of dealing with this problem. We have adopted the procedure of reducing the step size from Δa to $f\Delta a$, where $f < 1$, so that the refinement of a is $(a_0 - f\Delta a)$. We calculate the fit using a set of f values between 0 and 1 and then select that f which yields the minimal fit value. Although this procedure may not be as efficient as others in terms of computing time, it works well and precludes divergence. A listing of the program, which is written in CDC Extended Fortran, will be sent on request.

EXPERIMENTAL

Rhodizonic acid dipotassium salt was obtained from the Aldrich Chemical Company. The crude sample was purified by dissolution in an approximately stoichiometric quantity of dilute HCl solution, decantation to remove a small amount of insoluble matter, and precipitation by addition of KOH solution to pH ~ 6 . The process was repeated once more and a greenish-black substance isolated and used without further purification.

An Orion Model 801 pH meter equipped with a conventional glass and Ag,AgCl reference electrode was standardized against a 0.05 m potassium hydrogen phthalate buffer at $25.0 \pm 0.1^\circ C$ and pH 4.008. The standardization was repeated with each titration and was rechecked after the titration. In no case was more than 0.002 pH difference observed. Typical solutions of 50-mL volume containing about 0.2 mmol of rhodizonate were titrated with 0.1000 F HCl and were continuously purged by a stream of N_2 which had been saturated with water at ambient temperature (about $22^\circ C$) in order to avoid loss of solution volume. Finally, particular care was observed to ensure thermal and chemical equilibrium by allowing the meter reading to stabilize until no drift could be detected for a period of at least 3 min. This required standing times of about 15 to 30 min with each addition before the second end point.

To check the reliability of our experimental and calculational methods, we titrated two salts whose acid dissociation constants

Table I. Results of Calculation Based on Titration of Potassium Rhodizonate with HCl

	pK_1	pK_2	ν_{ep}	pH_{cal}
parameter values	4.377	4.661	2.053	-0.0014
variance coefficients	0.295	0.327	0.0544	0.0113
standard error based on variances s_r^2	0.0026	0.0028	0.0011	0.00052
standard error based on variances s_m^2	0.0050	0.0053	0.0022	0.00099

Degrees of freedom, 29. s_r^2 = variance of residuals, 0.236×10^{-4} . s_m^2 = variance of measurements, 0.863×10^{-4} . $\chi^2 = 7.9$.

are known with relatively high precision and used the calculational method described here to find pK values from these data. In one such experiment, a 80.0-mg sample of the dibasic salt sodium potassium *d*-tartrate- $4H_2O$ was titrated with HCl using an uncalibrated pH meter and recording 34 data points. Calculations done with these data yielded $pK_1 = 3.044 \pm 0.012$ and $pK_2 = 4.367 \pm 0.002$ to be compared with previously measured $pK_1 = 3.033 \pm 0.006$ and $pK_2 = 4.366 \pm 0.003$ reported by Bates and Canham (8). In a second experiment, sodium acetate titrated with HCl yielded $pK = 4.754 \pm 0.001$ for acetic acid, to be compared with a value of 4.756 measured by Harned and Ehlers (9). Values of the parameter pH_{cal} were calculated to be -0.184 ± 0.003 and -0.185 ± 0.001 from these experiments, whereas a direct measurement of the pH meter calibration offset with phthalate buffer was found to be -0.17 ± 0.02 .

RESULTS AND DISCUSSION

The three iterations described in the section titled "Interactive Digital Computer Program" were all well-behaved and converged easily to the results shown in Table I. We see that acid dissociation constants calculated are $pK_1 = 4.377 \pm 0.005$ and $pK_2 = 4.661 \pm 0.005$ where the uncertainties quoted are the estimated standard errors based on the more conservative variance estimates s_m^2 .

To check the reliability of these predictions from this single titration, we carried out two additional titrations. The procedures and calculations were similar to the one described above except that the pH meter was calibrated in the conventional way, data points were not recorded beyond the second equivalence point, and the least-squares fitting calculation involved only the three parameters pK_1 , pK_2 , and ν_{ep} . One titration yielded $pK_1 = 4.372$ and $pK_2 = 4.662$ from 21 recorded points, the other yielded $pK_1 = 4.391$ and $pK_2 = 4.655$ from 18 points. Grouping these three titrations, we find mean values and standard deviations of $pK_1 = 4.380 \pm 0.010$ and $pK_2 = 4.659 \pm 0.004$ which agree reasonably well with the parameter values and standard error estimates of the first titration.

We have mentioned earlier that the variance estimates of the parameters may be based on either the variance of residuals s_r^2 or on the effective variance of the measurements s_m^2 . In this experiment s_m^2 is itself based on our a priori estimate of the statistical uncertainties σ_y and σ_x in ν_d and pH_d , respectively, and are, therefore, rather crude. In principle σ_y and σ_x could be measured by performing other types of replication experiments to generate the statistics directly. The use of s_r^2 , on the other hand, avoids these troublesome replication experiments but, as discussed above, this variance is invalid if systematic error exists between the model equations and experimental "reality". Any significant systematic error introduces lack-of-fit which inflates the variance of residuals in an indefinite way and thus leads to unduly conservative parameter variances. In the titration whose calculations are summarized in Table I, we notice that s_r^2 turns out to be less than s_m^2 . This means (1) that our σ_y and σ_x estimates are a bit too large, and (2) that systematic error cannot be contributing significantly to the variance of re-

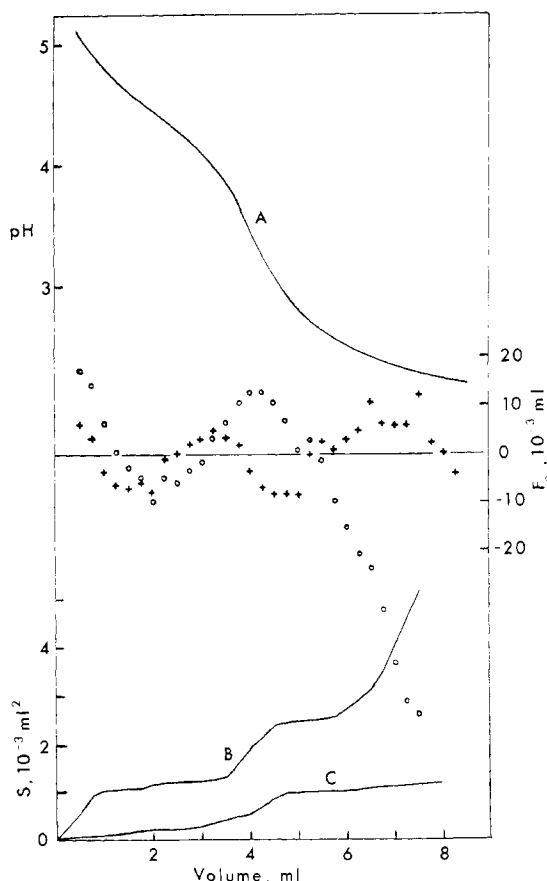


Figure 1. Curve A, pH vs. volume titration curve; function residuals F_0 from the calculation summarized in Table I (X) and from the calculation involving deliberately introduced systematic error and summarized in Table II (O); and cumulative sum of weighted residuals, S , for the Table I calculation, curve B, and the Table II calculation, curve C

siduals. These conclusions are summarized by the statistic χ^2 which is seen to be significantly less than the number of degrees of freedom. If s_m^2 were accurately known and lack-of-fit were absent, then χ^2 would be expected to average to the number of degrees of freedom.

Although this observation seems to rule out the existence of systematic error in this titration, it is advisable to check the pattern of residuals over the course of the titration. If the residuals are purely random, they will scatter about zero in a random way and exhibit no observable pattern. To illustrate this effect more clearly, we repeated the calculation and deliberately introduce a systematic error of reasonable magnitude. Rather than calculate the parameter pH_{cal} , which turned out to be -0.001 pH unit, we have fixed this constant at $+0.01$ pH unit. Considering that commercial buffer solutions are usually quoted to ± 0.02 pH unit, a pH calibration error of $+0.011$ is well within this tolerance if the meter were calibrated with such a buffer. The effect of this systematic error is immediately obvious from the results shown in Table II. The variance of residuals greatly exceeds the variance of measurements and $\chi^2 = 141$ for 30 degrees of freedom clearly reflects lack-of-fit. The pK values are each biased by ap-

Table II. Results of Calculation of Three Parameters with a Systematic Error Introduced by Fixing pH_{cal} at 0.01000

	pK_1	pK_2	ν_{ep}	pH_{cal}
parameter values	4.354	4.689	2.073	0.01000
variance coefficients	0.231	0.237	0.0184	
standard error based on s_r^2	0.0096	0.0098	0.0027	
standard error based on s_m^2	0.0044	0.0045	0.0013	

Degrees of freedom, 30. s_r^2 = variance of residuals, 4.02×10^{-4} . s_m^2 = variance of measurements, 0.856×10^{-4} . $\chi^2 = 141$.

proximately 0.02 unit when compared to the values shown in Table I and this bias is not adequately compensated by the statistical uncertainty estimates expressed as standard errors.

The pattern of residuals F_0 is shown in Figure 1 both for this calculation (O) and the more accurate calculation leading to the results in Table I (X). Also in Figure 1 are the cumulative sum of weighted residuals $\sum W_i F_{0i}^2$ as the titration progresses and the titration curve itself. We see that both sets of residuals exhibit oscillatory patterns. However, the amplitude of the set with systematic error is greater and the cumulative sum of weighted residuals is consequently greater for this case. The oscillation that is evident even when no systematic error is deliberately introduced indicates that some such error still exists although its effect is small in magnitude. Bard (7, p 202) notes that nonrandomness of residuals is a common occurrence when high precision data are fitted. The possible sources of systematic error are many: acidic or basic impurity may be present in the solid K_2R sample; the temperature may not be exactly 25°C and strictly invariant over the course of the titration; the pH meter may have drifted or may not have a strictly Nernstian response; the rhodizonic acid may have decomposed slightly during the titration, and so on. Presumably, further exploratory experimental work would lead us to the source of error and corrective measures could be taken to eliminate the source or to express it properly in a modified set of model equations. Then we would expect the oscillatory pattern of residuals to disappear, the variance of residuals to decrease, and the standard error estimates of the parameters based on this variance to decrease correspondingly. We did not carry out this extra experimentation since the increased precision in pK_1 and pK_2 that might result does not seem to be warranted.

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